LABORATORY · Lab · 26 hours

Experiment 10: Design and Implementation of an RC Lead Compensator



Control Systems Laboratory (22EEG53)

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LAB

Learning objectives

- Design an RC lead network for a required a and T
- Implement it and measure its frequency response
- Verify using MATLAB Bode plot
- Find the maximum phase lead

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

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Aim and apparatus

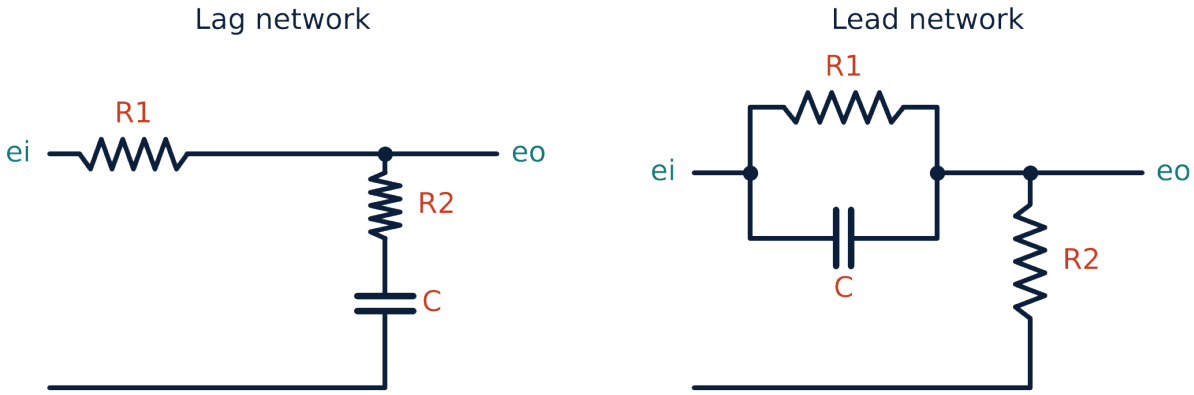
Aim: to design and implement an RC lead compensator and verify the result using MATLAB.

Apparatus: resistors, capacitor, function generator, CRO/DSO, bread board, MATLAB.

Design: a = 10 (φm ≈ 55°).

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RC lead network



Right: lead network

- Gc(s) = (1/a)(1 + aTs)/(1 + Ts).
- a = (R1 + R2)/R2; T = R1R2C/(R1 + R2).
- Zero at 1/(aT) = 1/(R1C); pole at 1/T.
- DC gain = R2/(R1 + R2) = 1/a.

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Design calculations

Problem Design a lead network with a = 10 using C = 0.1 μF and R2 = 10 kΩ.

- $a = (R1 + R2)/R2 = 10 \Rightarrow R1 = 9\ R2 = 90\ k\Omega$
- $T = R1R2C/(R1 + R2) = 9\ k\Omega \times 0.1\ \mu F = 0.9\ ms;\ \ aT = R1C = 9\ ms$
- $Gc = 0.1\ (1 + 0.009s)/(1 + 0.0009s)$
- $\varphi m = \sin^{-1}(9/11) = 54.9°$ at $\omega m = 1/(T\sqrt{a}) = 351\ rad/s$

Answer: R1 = 90 kΩ, R2 = 10 kΩ, C = 0.1 μF

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Procedure

1

Build

Assemble the lead network.

2

Excite

Sine input, 1 V peak.

3

Measure

Gain and phase from 100 to 5000 rad/s.

4

Plot

Bode plot; compare with MATLAB; verify φm and ωm.

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MATLAB verification

```
R1 = 90e3; R2 = 10e3; C = 0.1e-6;
a = (R1+R2)/R2; T = R1*R2*C/(R1+R2);
Gc = (1/a)*tf([a*T 1], [T 1])
bode(Gc), grid on
[mag, ph, w] = bode(Gc);
[phmax, i] = max(squeeze(ph));
fprintf('Max lead = %.1f deg at %.0f rad/s\n', phmax, w(i))
```

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Observation table

f (Hz)	ω (rad/s)	Vo (V)	Gain (dB)	Phase (deg)
20	126			
56	351			
200	1257			
800	5026			

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Key takeaways

- Lead: Gc = (1/a)(1 + aTs)/(1 + Ts), a = (R1 + R2)/R2, T = R1R2C/(R1 + R2)
- Maximum lead 54.9° at 351 rad/s for a = 10
- Passive network attenuates by 1/a: add gain a
- Experimental Bode matches MATLAB

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Practice questions

Q1. Design a lead network with a = 4 using C = 0.1 µF.

Q2. Compute φm and ωm of your design.

Q3. Why must the amplifier gain a be added?

Q4. Compare lead and lag networks.